# FIITJEE
# ALL INDIA TEST SERIES
## JEE (Advanced)-2023
## FULL TEST – III
## PAPER –2
## TEST DATE: 08-02-2023

# ANSWERS, HINTS & SOLUTIONS

## *Physics*

**PART – I**

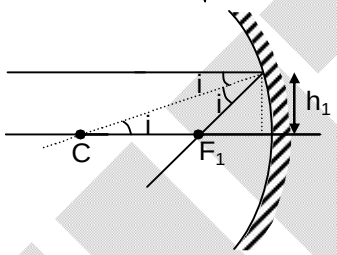
**Section – A**

1. B

Sol. Since B & revolve in opposite since, thus they are oppositely charged. From direction of motion of the charges $Q_A$ = + ve $Q_B$ & $Q_D \rightarrow$ -ve $Q_C \rightarrow$ neutral

2. A

Sol. $CF_1 = \dfrac{R}{2\cos i} = \dfrac{R^2}{2\sqrt{R^2 - h_1^2}} \text{ and } (h_1 = 3 \text{ cm})$




$CF_2 = \dfrac{R^2}{2\sqrt{R^2 - h_2^2}} \text{ and } (h_2 = 4 \text{ cm}) \quad \Rightarrow \quad CF_2 - CF_1 = \dfrac{25}{24} \text{ cm}$

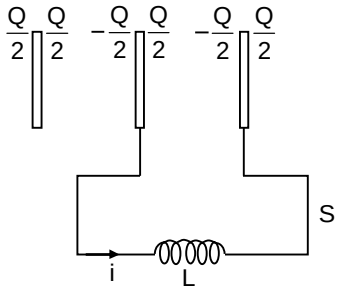
3. A

Sol. If maximum loss in kinetic energy is greater than first excitation energy inelastic collision may take place.

4. D

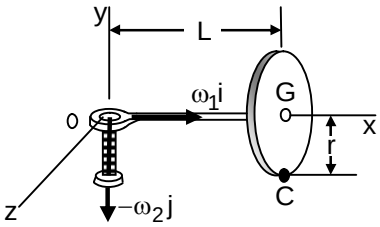
Sol. $\dfrac{\left(\dfrac{Q}{2}\right)^2}{2C} = \dfrac{1}{2}Li_0^2$

$\Rightarrow \; I_0 = \dfrac{Q}{2\sqrt{LC}}$



5. ACD

Sol. $\omega = \omega_1 i - \omega_2 j$




$v_C = \omega \times r_C = 0$

$(\omega_1 i - \omega_2 j) \times (Li - r j) = 0$

$(L\omega_2 - r\omega_1)k = 0$ $\qquad \omega_2 = \omega_1 i / L$

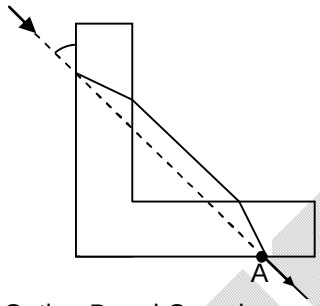
$\omega = \omega_1 i - (r\omega_1 / L) j$

$H_x = I_x \omega_x = \left(\frac{1}{2} m r^2\right) \omega_1$

$H_z = I_z \omega_z = \left(mL^2 + \frac{1}{4} m r^2\right) 0 = 0$

6. AD

Sol.




Option B and C are incorrect.

7. ABC

Sol. Let N be the number of fringes within the length x, then we have,

$\beta N = x \Rightarrow \frac{D\lambda}{d} N = x \Rightarrow N = \frac{xd}{\lambda D}$

At any time t

$N = \frac{x}{\lambda D}(d + vt)$

$\frac{dN}{dt} = \frac{xv}{\lambda D}$.

8. ABCD

Sol. $r = \frac{mV_0}{qB}$, and $PQ = 2r \sin\alpha = \frac{2mV_0}{qB} \sin\alpha$

$\alpha = \beta$

Time taken $(t) = \frac{2\pi r}{V}$

9. ABC

Sol. From graph in time from $t = 0$ to $t = 3$ sec.

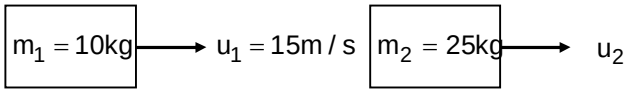
Acceleration of object of mass $m_1 = 10$ kg is

$$a = \frac{15-0}{3} = 5 \text{ m/s}^2$$

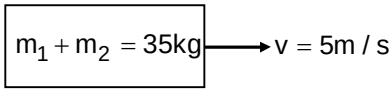
$\therefore$ Force on object of mass $m_1$ from $t = 0$ to $t = 3$ sec. ...(i)

$= 10 \times 5 = 50$ N

Before and after collision at $t = 4$ sec, the velocities of blocks are as shown.




Before collision




After collision

$\therefore$ Initial momentum of system

$= m_1u_1 + m_2u_2 = 150 + 25\ u_2$

Final momentum of system

$= (m_1 + m_2)v = 35 \times 5 = 175$

From conservation momentum

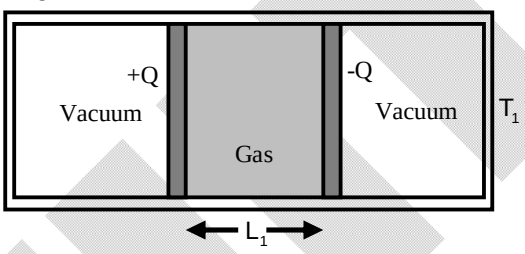
$\therefore\ 150 + 25\ u_2 = 175$

or $u_2 = +1$ m/s

$\therefore$ speed of second particle just before collision is 1 m/s and before collision both blocks move in same direction.

10. BCD

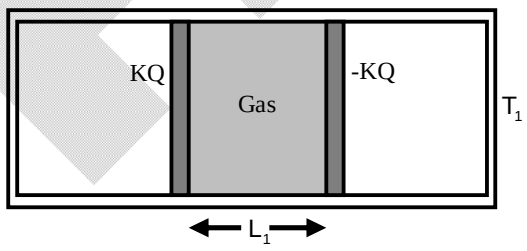
Sol. Stage 1




$P_1AL_1 = nRT_1$ $\quad \frac{Q^2}{2A\varepsilon_0} = P_1A = \frac{nRT_1}{L_1}$ (Force Balancing)

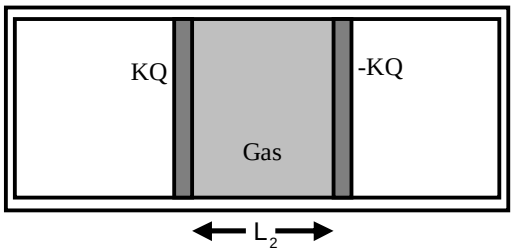
Stage 2

Charge is suddenly changed




System is not in equilibrium

Stage 3




$L_2CL_1$

Finally system in equilibrium

$P_2AL_2 = nRT_2 \quad \dfrac{K^2Q^2}{2A\varepsilon_0} = P_2A = \dfrac{nRT_2}{L_2}$

Energy of the system is conserved between stage 2 and 3.

$\Rightarrow \dfrac{1}{2}\varepsilon_0 E_2^2 AL_1 + \dfrac{f}{2}nRT_1 = \dfrac{1}{2}\varepsilon_0 E_2^2 AL_2 + \dfrac{f}{2}nRT_2$

On solving all the above equation

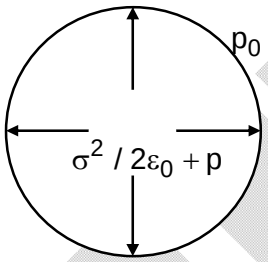
$\dfrac{T_2}{T_1} = \dfrac{2K^2 + f}{2 + f} \qquad w_{ext} = \left(K^2 - 1\right)\dfrac{Q^2L_1}{2A\varepsilon_0} = \left(K^2 - 1\right)nRT_1$

## Section – B

11. 6

Sol. $V = \dfrac{4}{3}\pi r^3$ .... (1)

$\sigma = \dfrac{Q}{4\pi r^2}$ ....(2)




$\dfrac{\sigma^2}{2\epsilon_0} + p - p_0 = \dfrac{4s}{r}$ .... (3)

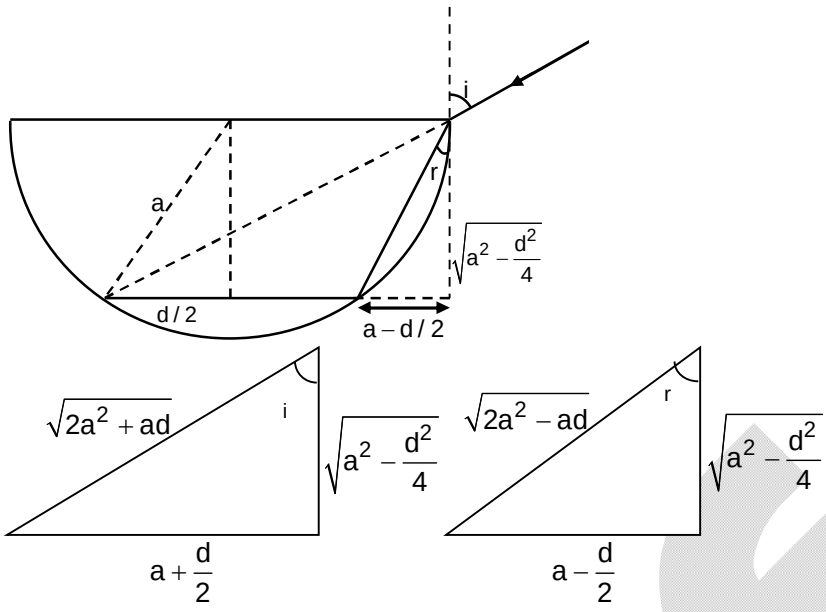
$p - p_0 = 0$ .... (4)

$\Rightarrow \quad \dfrac{1}{2\varepsilon_0} \cdot \dfrac{Q^2}{16\,\pi^2 r^4} = \dfrac{4s}{r}$

$\Rightarrow \quad n = 6$

12. 1

Sol.







$\sin i = \mu \sin r$

$\Rightarrow \quad \frac{d}{a} = 1$

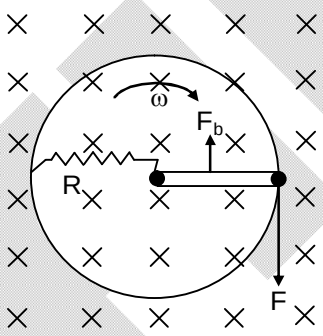
13. 2

Sol. $F_b = BIL$, $(\therefore L = r = 1m)$

Induced current: $I = \frac{\left(B\omega r^2/2\right)}{R}$

$\therefore \quad F_b = B\left(\frac{B\omega r^2}{2R}\right)r = \frac{B^2\omega r^3}{2R}$

To maintain constant angular velocity:

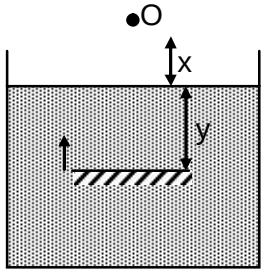



$F(r) = F_B(r/2)$

$\Rightarrow \quad F = \frac{F_B}{2} = \frac{B^2\omega r^3}{4R} = \frac{(10)^2 \times 2\pi \times (1)^3}{4 \times \pi} = 50N$

14. 3

Sol. Apparent distance of mirror from O




$= x + \frac{y}{\mu}$

Distance of final image from O

$= 2\left(x + \frac{y}{\mu}\right)$

Velocity of image

$= 2\left(\frac{dx}{dt} + \frac{1}{\mu}\frac{dy}{dt}\right) = \frac{2}{\mu} \times 8 = \frac{2}{4/3} \times 8 = 12 \text{ cm/s}$

15. 3

Sol. For $S_1S_2 = 2.5\lambda$ max path different $= 2.5\lambda$

Min path different $= 0$

Between $2.5\lambda$ and 0 lie $2\lambda$ and $\lambda \Rightarrow$ two circular bright fringes

$n_1 = 2$

For $S_1S_2 = 5.7\lambda$, max. path different $= 5.7\lambda$

Min path different $= 0$

Between $5.7\lambda$ and 0 lie $5\lambda$, $4\lambda$, $3\lambda$, $2\lambda$, $\lambda \Rightarrow$ Five circular bright fringes.

$\Rightarrow n_2 = 5$

$\therefore n_2 - n_1 = 5 - 2 = 3$

16. 9

Sol. $\sqrt{f} = a(Z-1)$

$\sqrt{f} = a(31-1)$

$\sqrt{\frac{25f}{x}} = a(51-1)$

$x = 9$

17. 5

Sol. $f_m = \frac{v}{(v - v_s\cos\theta)_{min}} \times f$

$1800 = \frac{v}{(v + v_s\cos\theta)_{max}} \times f$

On solving,

$f_m = 2250$ Hz

$= 2300 - 50$

18. 2

Sol. Net power given to $N_2$ gas $= 100 - 30 = 70$ cal/s

The nitrogen gas expands isobarically

$\therefore \frac{dq}{dt} = n\,Cp\,\frac{dT}{dt}$ or $70 = 5 \times \frac{7}{2}\,R\,\frac{dT}{dt}$

$\therefore \frac{dT}{dt} = 2k/\sec$

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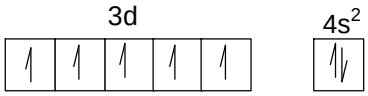
# Chemistry

## PART – II

### Section – A

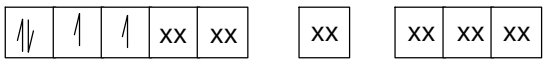
19. C

Sol. Mn




$[\overset{+3}{Mn}(CN)_6]^{3-}$ $\mu = 2.8$ so $n = 2$ unpaired

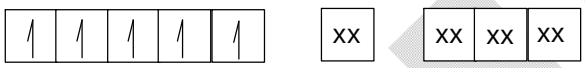
for $[\overset{3+}{Mn}(CN)_6]^{3-}$




Octahedral

$[\overset{2+}{Mn}Br_4]^{2-}$ $\mu = 5.5$ so $n = 5$

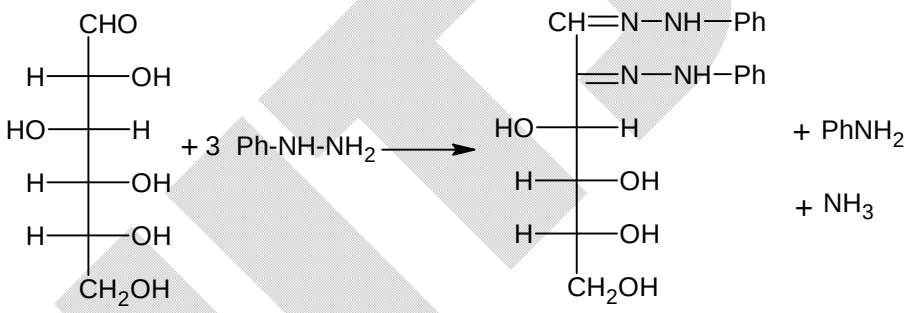
so,




Tetrahedral

20. A

Sol.




21. D

Sol. $3A \rightleftharpoons B$ $K_C = 8\times10^{15}$

| | | |
|---|---|---|
| Initially | 3 | 0 |
| Assume | 3-3 | 1 |
| | = 0 | 1 |
| Actually | 3x | 1-x |
| | | ≃ 1 |

So $\dfrac{1}{(3x)^3} = 8\times10^{15}$

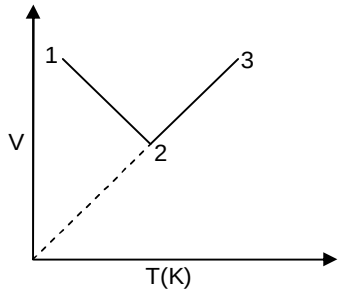
$(3x)^3 = \dfrac{1}{8\times10^{15}}$

$3x = \dfrac{1}{2}\times10^{-5}$

$[A] = 3x = 5\times10^{-6}$

22. B
Sol. Fact.

23. ABC
Sol.

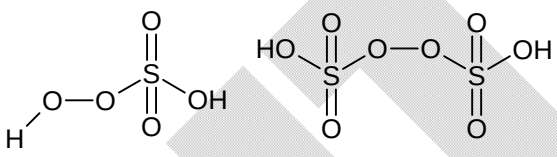



(A) for $1 \rightarrow 2$
$V \downarrow$ & $T \uparrow$ so
By $PV = nRT$

$$P = \frac{nRT}{V}$$

P will $\uparrow$
(B) for $2 \rightarrow 3$
$V \propto T$
So P will be constant
(C) at 1 & 3, T is higher at 3 so
$P_{(3)} > P_{(1)}$

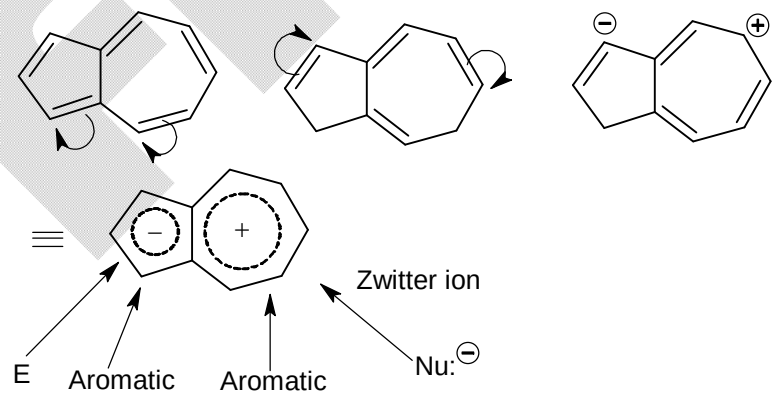
24. BD
Sol.



(B) (D)

25. ACD
Sol. Dacron is a condensation polymer. Teflon, neoprene and acrylonitrile are addition polymers. Hence option (A, C and D) is the answer.

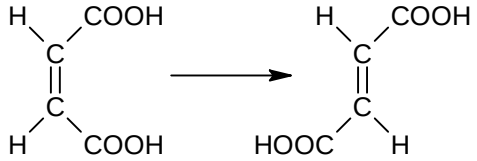
26. ABCD
Sol.



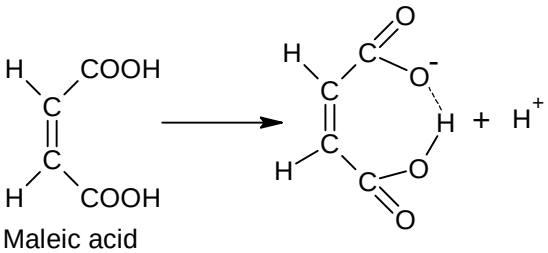
27. ABD

Sol. Maleic acid is stronger acid than fumaric acid

Maleic acid form cyclic ring after ionization while fumaric acid cannot do so that's why maleic acid $(pK_{a1} = 1.9,\ pK_{a1} = 6.1)$ is more stringer acid than fumaric acid $(pK_{a1} = 3.0,\ pK_{a1} = 4.4)$.



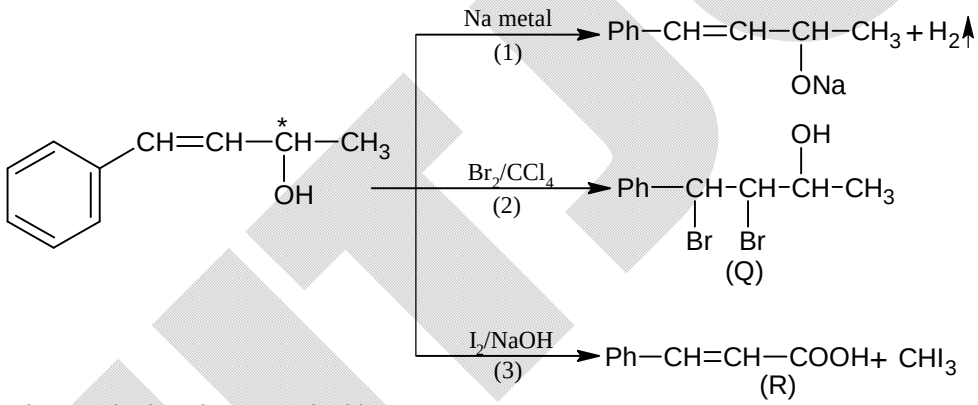
Maleic acid Fumaric acid



Maleic acid

28. BC

Sol.

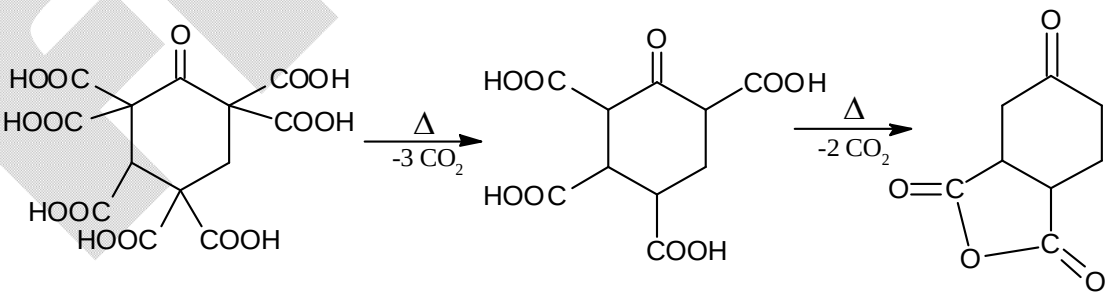


'P' has optical and geometrical isomers.

## Section – B

29. 5

Sol.



5 mole $CO_2$

30. 6

Sol. The following will release $H_2$ gas with propyne

$CH_3CH_2MgBr$ (III), NaH (IV), $NaNH_2$ (V), Na (VI), $\{(CH_3)CH\}_2NLi$ (VIII), $CH_3CH_2Li$ (IX)

31. 6

Sol. Glucose Fructose Mannose Maltose Cellobiose Lactose

32. 6

Sol. $Hg^{2+} + 2e^- \longrightarrow Hg \qquad \Delta G_1^0 = -2F(0.85)$

$Hg + 4I^- \longrightarrow HgI_4^{2-} + 2e^- \qquad \Delta G_2^0 = -2F(0.05)$

So

$Hg^{2+} + 4I^- \longrightarrow HgI_4^{2-} \qquad \Delta G^o = -2F(0.90)$

$0.90 = \frac{0.06}{2}\log K_f$

$K_f = 1.0 \times 10^{30}$

33. 3

Sol. m.m. of $NaOH = 30 \times 0.2 = 6$ m.m

m.m. of $CH_3COOH = 50 \times 0.2 = 10$ m.m

Now:

$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$

10 6 0

$10 - 6 = 4$ $6 - 6 = 0$ 6

$pH = -\log\left(2 \times 10^{-5}\right) + \log\frac{6}{4}$

$= 4.87$

Suppose 'v' mL of NaOH is added then

m.m of $CH_3COONa = (6 + v \times 0.2)$

m.m of $CH_3COOH = (4 - v \times 0.2)$

$5 = -\log\left(2 \times 10^{-5}\right) + \log\left(\frac{6 + 0.2v}{4 - 0.2v}\right)$

$0.3010 = \log\left(\frac{6 + 0.2v}{4 - 0.2v}\right)$

So $\left(\frac{6 + 0.2v}{4 - 0.2v}\right) = 2 \Rightarrow v = 3.33mL$

34. 8

Sol. % decrease in vol $= \frac{V_{bcc} - V_{fcc}}{V_{bcc}} \times 100$

$= \frac{\frac{1}{d_{bcc}} - \frac{1}{d_{fcc}}}{\frac{1}{d_{bcc}}} \times 100$

$$= \frac{\frac{N_A(4r)^3}{2\times 56\times 3\sqrt{3}} - \frac{N_A(4r)^3}{4\times 56\times 2\sqrt{2}}}{\frac{N_A(4r)^3}{2\times 56\times 3\sqrt{3}}}$$

$$= \frac{\frac{1}{3\sqrt{3}} - \frac{1}{2\times 2\sqrt{2}}}{\frac{1}{3\sqrt{3}}}$$

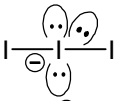
$$= \frac{0.46}{5.65}\times 100 \approx 8\%$$

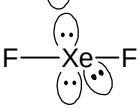
35. 7

Sol. $O = C = O$

$N \equiv \overset{+1}{N} - \overset{-2}{N}$

$H - C \equiv C - H$





$Cl - Be - Cl$

$S = C = S$

are linear

36. 2

Sol. $+\frac{1}{2}\frac{d[B]}{dt} = k[A]^n$

Taking ln on both side

$$\ln\left(\frac{1}{2}\right) + \ln\left(\frac{d[B]}{dt}\right) = \ln k + n\ln[A]$$

$$\ln\left(\frac{d[B]}{dt}\right) = \left(\ln k - \ln\left(\frac{1}{2}\right)\right) + n\ln[A]$$

So $n = 2$

# Mathematics

## PART – III

### Section – A

37. D

Sol. $\because 2(\alpha+\beta)=(\alpha-\beta)+\left(\frac{1}{\alpha}+\frac{1}{\beta}\right)$

$\Rightarrow \left(2-\frac{1}{\alpha\beta}\right)=\frac{\alpha-\beta}{\alpha+\beta}$

$\Rightarrow (2+a)=\frac{\alpha-\beta}{\alpha+\beta}>0 \Rightarrow a+2>0 \Rightarrow a>-2$

But $\alpha\beta<0 \Rightarrow \frac{a}{-1}<0 \Rightarrow a>0$

38. D

Sol. $\because \frac{(3^3-8)}{(3^3+8)}\times\frac{(4^3-8)}{(4^3+8)}\times\frac{(5^3-8)}{(5^3+8)}\times.....\times\frac{(n^3-8)}{(n^3+8)}$

Use: $n^3-8=(n-2)(n^2+2n+4)$

and $n^3+8=(n+2)(n^2-2n+4)$

We get, product $=\frac{1.2.3.4.(n^3-3)(n^2+2n+4)}{(n-1)n.(n+1)(n+2).7.12}$

For $n\to\infty$, its value $=\frac{1.2.3.4}{7.12}=\frac{2}{7}$

39. C

Sol. Let P $(t, t^2)$ and Q $(s, s^2)$.

So, equation of normal at P is $x+2+y=t+2t^3$

$\therefore$ Area $=\int_s^t\left(t^2+\frac{1}{2}-\frac{x}{2t}-x^2\right)dx=\frac{4}{3}\left(t+\frac{1}{4t}\right)^3\geq\frac{4}{3}$ $(AM\geq GM)$

(Here, take t > 0)

40. B

Sol. Let $1+\tan x=t^2$

$\frac{(t^2-1)^2}{(t^2-t)^2}<\frac{t^4+t^2+16}{t^4}$

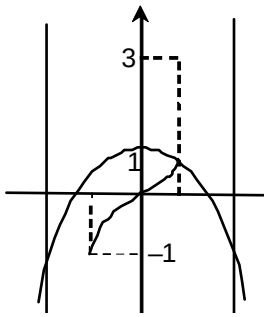
$\Rightarrow t^3<8 \Rightarrow t<2$

But $t>0$ and $t\neq 1$

$\Rightarrow \tan x\in(-1, 0)\cup(0, 3)$

$\Rightarrow -\frac{\pi}{4}<x<0$ or $0<x<\tan^{-1}3$

Only one solution on graph.



41. AC

Sol. $\int_0^{\pi} \frac{\sin^3 x dx}{x+1} = \int_0^{\pi} \underset{(I)}{\sin^3 x} \underset{(II)}{\frac{1}{(1+x)}} dx$

$= 0 - 3\int_0^{\pi} \underset{(I)}{\sin^2 x}.\underset{(II)}{\ln(x+1)} dx$

$= -3\left[0 - \int_0^{\pi} \sin 2x.(x+1)\{\ln(x+1)-1\} dx\right]$

& $\int_0^{\pi} \frac{\sin^3 x dx}{2x-\pi} = 2\int_0^{\pi/2} \frac{\sin^3 x}{2x-\pi} dx = 2\int_0^{\pi/2} \frac{\cos^3 x}{\pi-2x-\pi} dx$

$= -\int_0^{\pi/2} \frac{\cos^3 x}{x} dx$

42. ABC

Sol. $\because \alpha^2 - \alpha - 1 = 0 \Rightarrow \alpha^6 - \alpha^3 - 1 = 3.\alpha^3$

$\Rightarrow \alpha^6 - 1 = 4\alpha^3$

$\Rightarrow \frac{\alpha^6 - 1}{\alpha^3} = 4$

Similarly,

$\beta^2 - \beta - 1 = 0$

$\beta^6 - \beta^3 - 1 = 3.\beta^3$

$\frac{\beta^6 - \beta^3 - 1}{\beta^3} = 3$

43. BCD

Sol. $\because f(n) = (n+1)\ln\left(\frac{n+1}{n+2}\right) + \ln\left(\frac{n+1}{2}\right)$

$\Rightarrow g(n) = \ln\left(\frac{n+1}{2}\right) \Rightarrow g(x) = \log_e\left(\frac{x+1}{2}\right)$

$\Rightarrow g'(x) = \frac{2}{x+1}.\frac{1}{2} \Rightarrow g'(0) = 0$

$g''(x) = -\frac{1}{(x+1)^2} \Rightarrow g''(0) = -1$

& $\lim_{n\to\infty} \sum_{r=1}^{r=n} \left(\frac{1}{2.e^{\log_e\left(\frac{r+1}{2}\right)} - 1 + n}\right) = \sum_{r=1}^{r=n} \frac{1}{r+1-1+n} = \sum_{r=1}^{r=n} \frac{1}{n} \frac{1}{\left(\frac{r}{n}+1\right)}$

44. AD

Sol. $\because$ (2, 1) lies on circle

$\therefore (2-r)^2 + (1-r)^2 = r^2$ for touching circles on axis.

$\Rightarrow r^2 - 6r + 5 = 0$

$\Rightarrow r = 1$ or 5

45. BCD

Sol. (A) $\frac{f(x)}{x^2}=\frac{2}{3}\Rightarrow g(x)=3f(x)-2x^2$

$g(0)>0$ & $g(1)>0$ or $<0$

So, can't say.

(B) $g(x)=2f(x)=3x^2$

$\therefore g(0)>0$ & $g(1)<0$

Similarly, (C) and (D).

46. AB

Sol. Probability $=\frac{5^3+5-1}{5^4}=0.203$

## Section – B

47. 5

Sol. $y\geq 0$

$\Rightarrow \frac{e^y}{[y+1]}\leq 1$

$\Rightarrow \frac{2x^2+ax+2}{x^2+x+1}\geq 0$

$\Rightarrow x^2+x(a-1)+1\geq 0\forall x\in R$

$\Rightarrow -1\leq a\leq 3$

So, no. of integer a = 5

48. 7

Sol. $\because |A|=\alpha.\bar{\alpha}=|\alpha|^2$ and $|A-I|=\begin{vmatrix}\alpha-1 & 0\\ 2 & \bar{\alpha}-1\end{vmatrix}$

$|A-I|=\alpha\bar{\alpha}-\alpha-\bar{\alpha}+1=|\alpha-1|^2$

$\Rightarrow |A-I|=|\alpha|^2-tr(A)+1$

Given, $|2A^2-2A|=144$

$\Rightarrow |A^2-A|=36$

$|A||A-I|=36$

$\Rightarrow |\alpha|^2|\alpha-1|^2=36$

$|\alpha(\alpha-1)|=6$ or $-6$

$\alpha^2-\alpha=6$ or $-6$

$\alpha=3$ or $-2$ or $\alpha=\frac{+1\pm i\sqrt{2}.3}{2\times 1}$

$\bar{\alpha}=3$ or $-2$ or $\bar{\alpha}=\frac{+1\mp i\sqrt{2}.3}{2\times 1}$

$\alpha+\bar{\alpha}=6$ or $-4$ or $1$

49. 8

Sol. $\because \quad \beta = \dfrac{f(4)+f(12)}{2}$

$= \dfrac{4+12}{2} = 8$

50. 4

Sol. $f^{-1}(x) = x-1$ then $h^{-1}(x) = \left(g \cdot f(x)\right)^{-1}$

$\therefore \dfrac{d}{dx}\left(h^{-1}(x)\right) = 3x^2+1 \quad = f^{-1} \cdot g^{-1}(x)$ (reversal law)

$= f^{-1}\left(x^3+x^3+x+1\right)$

$= x^3+x$

$\left(h^{-1}(1)\right)' = 3+1 = 4$

51. 6

Sol. $|\alpha+\beta| \leq |\alpha|+|\beta|$

Now,

$i\alpha^3+\alpha^2-\alpha^2-\alpha+i=0$

$\Rightarrow \alpha^2(i\alpha+1)+i(i\alpha+1)=0$

$\Rightarrow |\alpha| = 1$

so $|\alpha+\beta| \leq 1+5 \leq 6$

52. 1

Sol. Given that $A^{-1}, T = A = (A^T)^{-1} = AA^T = I$

$\Rightarrow |A| = \pm 1$

Now, $|\text{adj}(\text{adj}A)| \;=\; |A|^{2^2} = 1$

53. 4

Sol. $p \in \left[\dfrac{3}{2},2\right], q \in \left[\dfrac{1}{4},1\right]$

$\Rightarrow \dfrac{7}{4} \leq p+q \leq 3$

$\Rightarrow \left[(p+q)_{max}\right] + \left[(p+q)_{min}\right]$

$= 3+1 = 4$

54. 2

Sol. $n_1 = 2(7+6+5)(7!) = 36(7!)$

$n_2 = 16(7!)$

$\left[\dfrac{n_1}{n_2}\right] = \left[\dfrac{36}{16}\right] = 2$